T3 group factors: UV

Active theory: SM + S_1 + S_2 + F. The same beta function is shown across runs in a single comparison table, with exact analytic group factors in the run columns.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Field representations

run	S_1	S_2	F	α
T3 − E	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	-4

Gauge and Yukawa beta functions

$\beta_{g_1}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
g_1^3	$\frac{179}{6}$

$\beta_{g_2}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
g_2^3	$\frac{1}{6}$

$\beta_{g_3}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
g_3^3	-7

$\beta_{Y_u}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1

$\beta_{Y_d}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1

$\beta_{Y_e}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{3}{4}$
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1

$\beta_{y_1}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$T_\nu^{(1)}(y_1)_{ij}$	1
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{3}{2}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{2}$
$g_1^2(y_1)_{ij}$	$-\frac{15}{2}$

$\beta_{y_2}^{(1)}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$T_\nu^{(2)}(y_2)_{ij}$	1
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{3}{2}$
$g_2^2(y_2)_{ij}$	$-\frac{9}{2}$
$g_1^2(y_2)_{ij}$	$-\frac{15}{2}$

β_{M_F}

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$(y_1^T y_1^*)M_F$	$\frac{3}{4}$
$M_F(y_2^\dagger y_2)$	$\frac{3}{4}$
$g_2^2 M_F$	$-\frac{9}{2}$
$g_1^2 M_F$	$-\frac{27}{2}$

$\beta_{m_1^2}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$m_1^2\, T_\nu^{(1)}$	2
$M_F^\dagger M_F\, T_\nu^{(1)}$	-4
$\lambda_{S_1}^{(1)}\, m_1^2$	8
$g_2^2\, m_1^2$	-12
$g_1^2\, m_1^2$	-24
$\lambda_{12}^{(1)}\, m_2^2$	6
$\lambda_{S_1}^{(A)}\, m_1^2$	4
$\lambda_{12}^{(\times)}\, m_2^2$	4

$\beta_{m_2^2}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$m_2^2\, T_\nu^{(2)}$	2
$M_F^\dagger M_F\, T_\nu^{(2)}$	-4
$\lambda_{S_2}^{(1)}\, m_2^2$	8
$g_2^2\, m_2^2$	-12
$g_1^2\, m_2^2$	-6

$\lambda_{12}^{(1)}\, m_1^2$	6
$\lambda_{S_2}^{(A)}\, m_2^2$	4
$\lambda_{12}^{(\times)}\, m_1^2$	4

$\beta_{\lambda_{HS_1}^{(1)}}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$\lambda_{HS_1}^{(1)}\, T_\nu^{(1)}$	2
$\mathrm{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$	-2
$\lvert\lambda_5\rvert^2$	4
$g_2^2\, \lambda_{HS_1}^{(1)}$	$-\frac{33}{2}$
$g_1^2\, \lambda_{HS_1}^{(1)}$	$-\frac{51}{2}$
$\lambda_1\, \lambda_{HS_1}^{(1)}$	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4
$\lambda_{HS_1}^{(1)}\, \lambda_{S_1}^{(1)}$	8
$\lambda_{12}^{(1)}\, \lambda_{HS_2}^{(1)}$	6
g_2^4	6
g_1^4	12
$\left(\lambda_{HS_1}^{(A)}\right)^2$	2
$\lambda_{HS_1}^{(1)}\, \lambda_{S_1}^{(A)}$	4
$\lambda_{HS_2}^{(1)}\, \lambda_{12}^{(\times)}$	4
$\lambda_{HS_1}^{(1)}\, T$	2

$\beta_{\lambda_{HS_2}^{(1)}}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$\lambda_{HS_2}^{(1)}\, T_\nu^{(2)}$	2
$\mathrm{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$	-2
$\lvert\lambda_5\rvert^2$	4
$g_2^2\, \lambda_{HS_2}^{(1)}$	$-\frac{33}{2}$
$g_1^2\, \lambda_{HS_2}^{(1)}$	$-\frac{15}{2}$
$\lambda_1\, \lambda_{HS_2}^{(1)}$	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4
$\lambda_{HS_2}^{(1)}\, \lambda_{S_2}^{(1)}$	8
$\lambda_{12}^{(1)}\, \lambda_{HS_1}^{(1)}$	6
g_2^4	6
g_1^4	3
$\left(\lambda_{HS_2}^{(A)}\right)^2$	2
$\lambda_{HS_2}^{(1)}\, \lambda_{S_2}^{(A)}$	4
$\lambda_{HS_1}^{(1)}\, \lambda_{12}^{(\times)}$	4
$\lambda_{HS_2}^{(1)}\, T$	2

$\beta_{\lambda_{12}^{(1)}}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$\mathrm{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$	-2
$\lvert\lambda_5\rvert^2$	$\frac{4}{3}$
$g_2^2\, \lambda_{12}^{(1)}$	-24

$g_1^2\,\lambda_{12}^{(1)}$	-30
$\left(\lambda_{12}^{(1)}\right)^2$	4
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(1)}$	8
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(1)}$	8
$\lambda_{HS_1}^{(1)}\,\lambda_{HS_2}^{(1)}$	4
g_2^4	12
g_1^4	48
$\left(\lambda_{12}^{(A)}\right)^2$	4
$\lambda_{12}^{(1)}\,\lambda_{S_1}^{(A)}$	4
$\lambda_{12}^{(1)}\,\lambda_{S_2}^{(A)}$	4
$\left(\lambda_{12}^{(\times)}\right)^2$	4
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	4
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	4
$\lambda_{12}^{(1)}\,T_\nu$	2

β_{λ_5}

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$\lambda_5\,\lambda_1$	2
$\lambda_5\,\lambda_{HS_1}^{(1)}$	4
$\lambda_5\,\lambda_{HS_2}^{(1)}$	4
$\lambda_5\,\lambda_{12}^{(1)}$	2
$g_2^2\,\lambda_5$	$-\frac{33}{2}$
$g_1^2\,\lambda_5$	$-\frac{33}{2}$
$\lambda_5\,\lambda_{HS_1}^{(A)}$	-2
$\lambda_5\,\lambda_{HS_2}^{(A)}$	2
$\lambda_5\,\lambda_{12}^{(A)}$	2
$\lambda_5\,\lambda_{12}^{(\times)}$	-2
$\lambda_5\,T$	2
$\lambda_5\,T_\nu$	1

Additional scalar-coupling beta functions

$\beta_{\lambda_{HS_1}^{(A)}}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$g_1^2 g_2^2$	-24

$\beta_{\lambda_{HS_2}^{(A)}}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
$g_1^2 g_2^2$	-12

$\beta_{\lambda_{12}^{(A)}}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$

$g_1^2 g_2^2$	48

$\beta_{\lambda_{12}^{(\times)}}$

$\mathcal{S}$	$\mathrm{T3-E}\ (\alpha=-4)$
g_2^4	6
$\left(\lambda_{12}^{(A)}\right)^2$	2
$\lambda_{12}^{(1)}\,\lambda_{12}^{(\times)}$	8
$\left(\lambda_{12}^{(\times)}\right)^2$	10
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	-2
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	-2

First threshold

The first EFT GroupFactors report below is selected from the actual threshold plan stored in the run. No assumption is made here about whether F, S_1, S_2, or a degenerate set is integrated out first.